

Review Article

A systematic review of deep learning techniques for generating synthetic CT images from MRI data

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Abstract

Introduction: This systematic review evaluates various studies on deep learning algorithms for generating synthetic CT images from MRI data, focusing on challenges in image quality and accuracy in current synthetic CT generation methods. Magnetic resonance imaging (MRI) is increasingly important in clinical settings due to its detailed visualization and non-invasive nature, making it a valuable tool for advancing patient care and identifying new areas for research.

Materials and Methods: In this study we conducted a thorough search across several databases to identify studies published between January 2009 and January 2024 on using deep learning to generate synthetic CT (sCT) images from MRI for radiotherapy. The review focused on peer-reviewed, English-language studies and excluded unpublished, non-English, and irrelevant studies. Data on deep learning methods, input modalities, and anatomical sites were extracted and analyzed using a result-based synthesis approach. The review categorized 84 studies by anatomical site, following PRISMA guidelines for summarizing the findings.

Results: The U-Net model is the most frequently used deep learning model for generating synthetic CT images from MRI data, with 34 articles highlighting its effectiveness in capturing fine details. Conditional GANs are also widely used, while Cycle-GANs and Pix2pix are effective in image translation tasks. Significant differences in performance metrics, such as MAE and PSNR, were observed across anatomical regions and models, highlighting the variability in accuracy among different deep learning approaches.

Conclusion: This review underscores the need for continued refinement and standardization in deep learning approaches for medical imaging to address variability in performance metrics across anatomical regions and models.

Keywords: deep learning algorithms; synthetic CT; magnetic resonance imaging; radiotherapy planning; image quality; non-invasive imaging.

Introduction

In recent years, the integration of deep learning algorithms has revolutionized the practice of medical imaging and radiotherapy. Their ability to perform tasks such as image segmentation has significantly improved target localization and treatment planning.¹ These algorithms play a pivotal role in automating disease detection and diagnosis,² providing rapid and precise identification of abnormalities in medical images³. The technology of image registration techniques has been improved by a variety of deep learning models, which have facilitated the accurate diagnosis and contouring of tumors and organs at risk in radiotherapy.⁴ Additionally, deep learning has made a

substantial contribution to the art of image reconstruction by reducing noise and improving the quality of images in Computerized Tomography (CT) and Magnetic Resonance Imaging (MRI) scans.⁵ Radiomics and feature extraction find new dimensions with the help of deep learning, offering quantitative insights for disease characterization and treatment response prediction. Computer-Aided Diagnosis (CAD) systems, powered by deep learning, assist radiologists by automating analyses and highlighting potential anomalies.⁶ The application of deep learning extends to personalized treatment planning, where models leverage patient-specific characteristics for tailored approaches. Furthermore, these algorithms contribute to image enhancement, virtual and augmented reality

applications, and predictive modeling for patient outcomes.⁷ The integration of deep learning with genomic data provides a comprehensive understanding of diseases, influencing treatment decisions.⁸ Automated reporting, quality assurance, real-time image analysis during procedures, and drug discovery are additional areas where deep learning is shaping the future of medical imaging and radiotherapy, promising improved patient care and treatment efficacy. One area where these algorithms have shown great promise is in the generation of synthetic CT (sCT) (pCT) images from magnetic resonance imaging (MRI) data. Synthetic CT (sCT) images play a crucial role in various clinical applications such as radiation therapy planning where the absence of CT data can limit accurate treatment planning and in μ -map generation for attenuation correction in PET/MRI.

Magnetic Resonance Imaging (MRI) is increasingly important in clinical settings due to its non-invasive and detailed visualization capabilities, providing crucial insights into anatomical structures and pathological conditions.⁹ The study on generating synthetic CT (sCT) from MRI is crucial due to the growing reliance on MRI scans in diagnostic and treatment planning. Deep learning algorithms play a pivotal role in medical image processing, offering sophisticated tools for image analysis and feature extraction.¹⁰ Deep learning for generating synthetic CT (sCT) from MRI offers automation and efficiency, but challenges like model interpretability and generalizability necessitate a comprehensive understanding of its potential and limitations. In the context of radiotherapy, MRI plays an increasingly critical role, particularly in adaptive radiotherapy, where its non-ionizing nature and superior soft-tissue contrast offer advantages over traditional CT. Applications such as Cone Beam CT (CBCT) and MR-guided linear accelerators (MR-Linac) leverage MRI for precise targeting and dose planning in real-time, underscoring the value of synthetic CT (sCT) generation for comprehensive MRI-only workflows. This review critically analyzes studies on synthetic CT (sCT) generation from MRI using deep learning algorithms, highlighting challenges and comparative evaluations. It offers insights for clinical practice and treatment procedures. The review may guide future research and advance medical imaging technologies by identifying areas for exploration and refining synthetic CT (sCT) images from MRI data, contributing to more efficient imaging methodologies.

Recent Advancement in Synthetic CT (sCT) Generation

MRI does not inherently provide Hounsfield Unit (HU) or Electron Density (ED) values, which are essential for accurate dose calculations in radiotherapy. Consequently, synthetic CT (sCT) images are generated from MRI data to enable comprehensive MRI-only workflows in radiotherapy planning. To address this challenge, several sophisticated techniques have been developed for generating synthetic-CT (sCT) images from

MRI, which can be categorized into five primary approaches: bulk density overrides, voxel-based methods, patch-based techniques, atlas-based strategies, and learning-based models.¹¹ Bulk density overrides use fixed density values for different tissues, while voxel-based methods assign Hounsfield units at each voxel based on MRI intensity. Patch-based techniques match small MRI patches to MRI-CT pairs, enhancing local detail, and atlas-based strategies align MRI scans to a CT atlas using anatomical landmarks. Finally, learning-based models, such as those employing deep learning algorithms, offer advanced mapping from MRI to synthetic CT, providing high accuracy and flexibility for complex anatomical structures. While some research is dedicated to the application of sCT generation in MRI-only radiation therapy focusing on the accuracy of dose calculation, other studies take a broader perspective, evaluating the overall image quality and versatility of sCT volumes across various clinical scenarios.^{12,13}

Differences between Traditional Methods Vs. Deep Learning Approaches

The generation of synthetic CT (sCT) (pCT) images from MRI data involves a comparison between traditional methods and deep learning approaches. Traditional methods for pCT generation often rely on segmentation-based, intensity-based, atlas-based, or hybrid techniques to map MRI data to CT-like images.^{14,15} These methods typically involve manual feature extraction and predefined algorithms to approximate CT numbers from MRI intensities. Conversely, the utilization of convolutional neural networks has enabled the emergence of deep learning methodologies as potent instruments for the generation of pCTs (CNNs).^{16,17} The automatic learning of the mapping between MRI and CT data has been demonstrated to be superior by deep learning methods, including U-Net, GANs, and ResNet.^{16,18} This eliminates the necessity for handcrafted features and improves accuracy. The U-Net architecture is a convolutional neural network (CNN) designed specifically for biomedical image segmentation. Its symmetric encoder-decoder structure with skip connections preserves spatial information, making it effective for detailed imaging tasks. Generative Adversarial Networks (GANs) consist of a generator and a discriminator working in opposition to generate realistic images; they are widely used for image synthesis and enhancement. ResNet, or Residual Network, is a deep CNN that uses skip connections to address the vanishing gradient problem, allowing for efficient training of very deep networks and improving model performance on complex image reconstruction tasks. These deep learning architectures can efficiently generate pCT images with reduced noise and improved resemblance to ground truth CT images, as demonstrated in studies comparing different deep learning models for pCT synthesis.¹⁶ It is worth noting that traditional methods for generating synthetic CT images remain widely used in clinical routines on MR-linacs. These methods

are well-established in practice due to their reliability and compatibility with existing workflows, despite recent advancements in deep learning approaches. However, deep learning approaches offer advantages in terms of accuracy, efficiency, and the capacity to manage complex relationships within the data.¹⁹ Deep learning techniques are being integrated into medical imaging, particularly for generating pCT images from MRI data, which could significantly improve treatment planning and other medical applications.²⁰

Methodology

This study employed a systematic literature review to gather and analyze relevant research on the use of deep learning algorithms for generating synthetic CT images from MRI data, specifically within the context of radiotherapy dose planning.²¹ The review incorporated studies across various evidence types, including qualitative, quantitative, and computational approaches. To ensure the robustness and reliability of the review, key principles such as transparency, clarity, comprehensiveness, and focus were adhered to throughout the process, minimizing potential biases in selection, publication, and data extraction.^{22,23}

The methodology was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, which were instrumental in structuring and summarizing the results of the literature search, as depicted in **Figure 1**.²⁴ This systematic approach allowed for a thorough exploration and synthesis of the existing body of knowledge, providing a comprehensive understanding of the advancements in synthetic CT (sCT) generation using deep learning techniques.

Inclusion and Exclusion Criteria

In this systematic review, we included peer-reviewed primary studies published in English from January 2009 to January 2024 that focused on generating synthetic CT (sCT) images from MRI using deep learning algorithms, specifically in the context of radiotherapy dose planning. We excluded unpublished research, non-English publications, literature reviews, and studies that did not use deep learning techniques or were not related to radiotherapy.²⁵ Our inclusion criteria ensured that the studies provided detailed data on deep learning architectures, loss functions, MR/CT types, input modalities, body regions of interest, patient cohort size, and results.

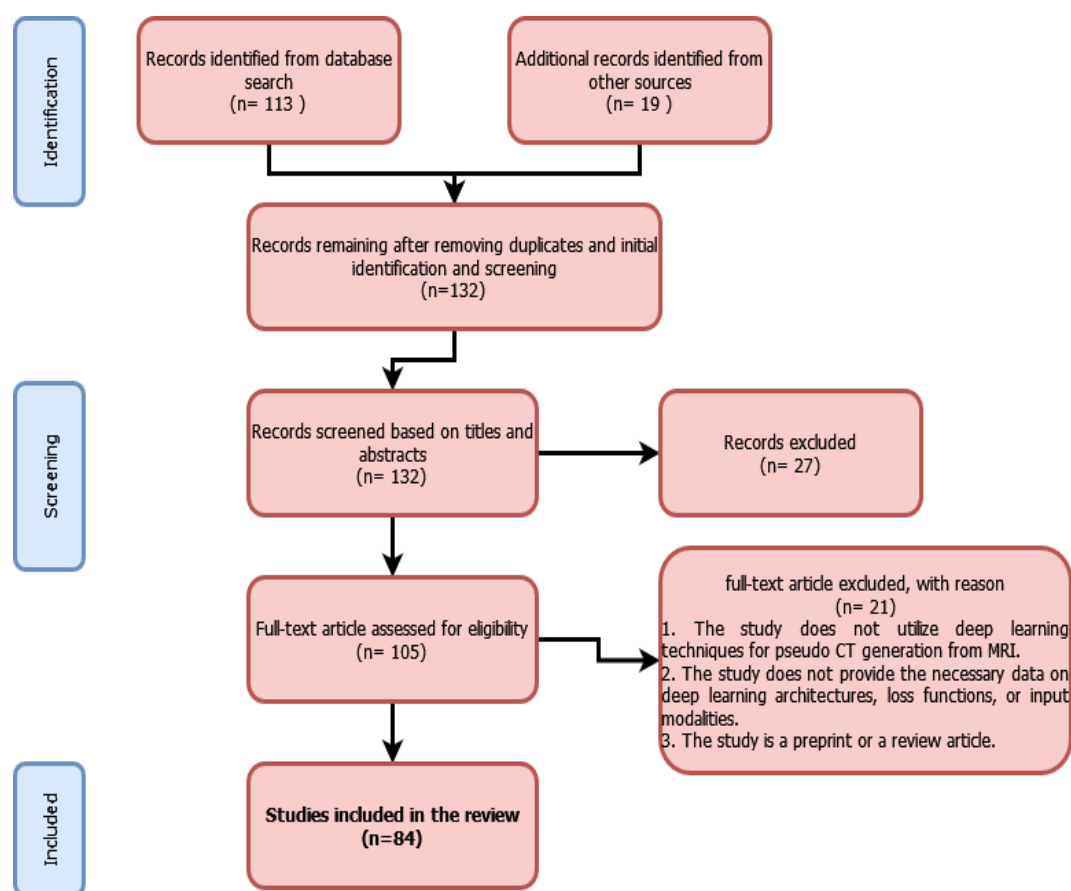


Figure 1. A PRISMA flowchart illustrating the process of study selection for the systematic review

Databases and Search Strategy

A systematic search was conducted across multiple databases to ensure a comprehensive collection of relevant studies. The databases searched included PubMed, Google Scholar, Science Direct, and general Google searches. The search strategy was designed to capture studies related to the generation of synthetic CT from MRI using deep learning algorithms, with a specific focus on radiotherapy dose planning. To refine the search, a set of carefully selected keywords was employed, including 'deep learning,' 'synthetic CT,' 'synthetic CT (sCT),' 'substitute CT,' 'MRI,' and 'radiotherapy.' These keywords were combined using Boolean operators (AND/OR) to increase the sensitivity and specificity of the search.^{24,25} The search was not restricted to these databases alone; an extensive internet search was also conducted to capture any additional relevant studies that might have been missed.

To ensure a thorough and exhaustive search, reference lists of identified articles were also reviewed to find additional studies. The entire search process adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, which are detailed in **Figure 1**.²⁴

Selection and Data Extraction

After the initial search, all relevant articles were exported to EndNote for organization and duplicate removal. Given the diverse methodologies employed in the studies, a thorough quality assessment was performed using a structured approach. Although quality scores were not used as exclusion criteria, they provided valuable insights into the reliability of the findings.^{24,26} For data extraction, a comprehensive template was developed, utilizing three tables to categorize and summarize the key information from each study. Details on the algorithms used were also extracted and synthesized in the results section to provide a comprehensive overview of deep learning applications

in synthetic CT generation. **Table 1** focused on the following metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Normalized Cross-Correlation (NCC). **Table 2** detailed the Dice Similarity Coefficient for each article, while **Table 3** captured data related to Dose Differences and Gamma Pass Rates. The extraction process was conducted systematically to ensure accuracy and consistency, with all relevant data meticulously recorded and reviewed.

Data analysis and synthesis

The study used a result-based convergent data synthesis approach to integrate findings from various studies, combining similar designs. A narrative synthesis was applied, analyzing content, categorizing comments, and identifying recurring themes. The studies retrieved were further analyzed based on several key criteria, including the type of deep learning architecture used, loss functions, MR/CT types and input modalities, body regions of interest, number of patients, and the outcomes of the studies.

Results

The results section begins with an overview of the deep learning models and architectures used in synthetic CT generation, followed by quantitative analyses presented in **Tables 1, 2, and 3**. This structure is intended to provide both a qualitative and quantitative synthesis of findings. The final analysis included 84 articles, categorized by anatomical site as follows: Brain (37 articles), Head and Neck (17 articles), Pelvis (12 articles), Prostate (10 articles), Abdomen (7 articles), Breast (3 articles), Liver (2 articles), Rectum (2 articles), Cervix (2 articles), and Kidney (1 article), as illustrated in **Figure 2**.

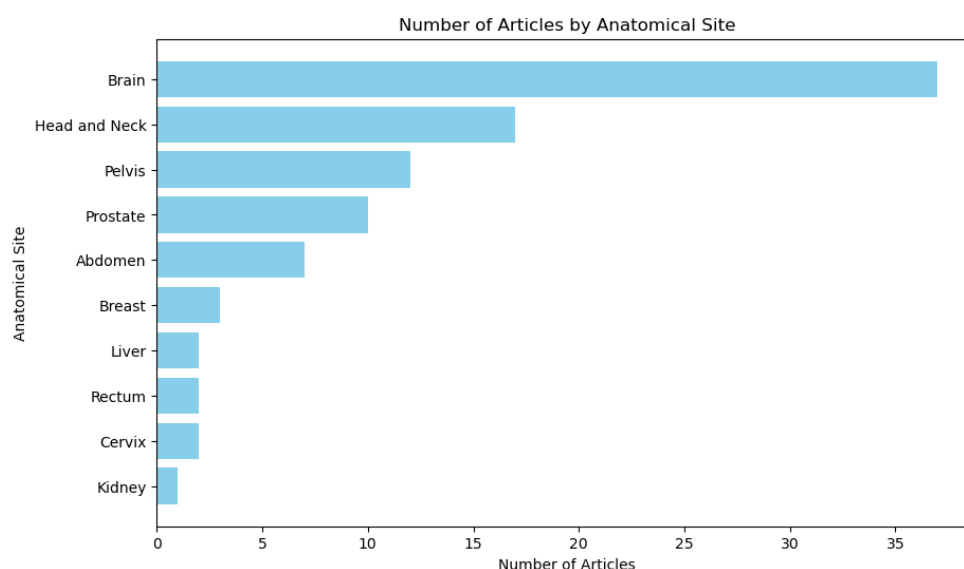


Figure 2. Visualization showing the number of articles categorized by anatomical site.

Deep learning models and algorithms

U-Net

U-Net is a convolutional neural network architecture designed specifically for biomedical image segmentation. It features a symmetric encoder-decoder structure with skip connections that help in preserving spatial information. U-Net has been widely adopted in synthetic CT (sCT) generation studies due to its effectiveness in capturing fine details and structures in medical images. For instance, Han (2017) utilized a 2D U-Net with an L1 loss function to generate synthetic CT (sCT) images, demonstrating its capability to produce accurate image reconstructions.²⁷ Similarly, Gupta et al. (2019) employed a 2D U-Net with a combination of Mean Absolute Error (MAE) and Mean Squared Error (MSE) losses, highlighting the model's flexibility in handling different loss functions to improve performance.²⁸ Nepl et al. (2019) further explored both 2D and 3D U-Nets, showcasing the model's adaptability to various training dimensions and imaging contexts.²⁹

Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) revolutionize image synthesis by combining a generator and discriminator to generate realistic images and enhance results through adversarial training. Nie et al. (2017) combined L2, gradient, and adversarial losses in a GAN with a fully convolutional network (FCN) generator and a CNN discriminator to enhance the quality of synthetic CT (sCT) images.³⁰ Emami et al (2018) utilized L1 and adversarial losses in a GAN framework with a ResNet generator and a CNN discriminator to create high-quality images.³¹ Kazemifar et al. (2019) illustrated the model's capacity to incorporate supplementary information to enhance image generation by employing a conditional GAN (cGAN) with binary cross-entropy and mutual information loss for the discriminator.³²

Cycle-GAN

Cycle-GAN enhances GANs by incorporating cycle consistency loss, allowing unpaired image-to-image translation, ensuring the translated image can be remapped to the original, preserving consistency between source and target domains. Wolterink et al. (2017) employed Cycle-GAN with least squares, adversarial, gradient difference, and cycle consistency losses to generate synthetic CT (sCT) images from MRI data.³³ Yang et al. (2018) also utilized Cycle-GAN, integrating adversarial, cycle consistency, and structure consistency losses to ensure high-quality image translation.³⁴ The ability of Cycle-GAN to handle unpaired data makes it particularly useful in scenarios where paired training datasets are not readily available.

Conditional GAN (cGAN)

Conditional GANs (cGANs) are a type of GAN where the generator and discriminator are influenced by additional information, such as class labels or data from another modality. This conditioning improves control over the generated images,

making cGANs particularly effective for medical image synthesis. In order to improve the quality of the generated images, Kazemifar et al. (2019) implemented cGANs for synthetic CT (sCT) generation, employing mutual information as the loss function.³² Koike et al. (2019) illustrated the model's efficacy in enhancing image generation by utilizing conditional inputs by employing cGANs with adversarial and L1 losses for 2D training.³⁵

Dilated CNN

Dilated Convolutional Neural Networks (Dilated CNNs) utilize dilated convolutions to expand the receptive field without losing resolution, allowing the network to capture more contextual information. Dinkla et al. (2018) applied a dilated CNN for 2.5D training with an L1 loss function, showcasing its ability to balance the trade-off between receptive field size and computational efficiency.³⁶ The use of dilated convolutions helps in capturing broader contextual information, which is crucial for accurate synthetic CT (sCT) generation.

Deep Convolutional Encoder-Decoder Network (DECNN)

Encoder-decoder networks are designed to learn efficient representations and reconstruct input data, making them suitable for image translation tasks. Xiang et al. (2018) utilized a Deep Convolutional Encoder-Decoder Network (DECNN) with L2 norm and embedding blocks for 3D training, demonstrating its effectiveness in detailed image reconstruction.³⁷ The encoder-decoder structure helps in capturing and reconstructing complex anatomical features, which is essential for generating accurate synthetic CT (sCT) images from MRI data.

Training methodologies

The analysis indicates that 2D training methodologies are the most commonly employed approach, used in 64 studies. This could be attributed to the lower computational requirements and relative simplicity of 2D models in comparison to 3D models. Nevertheless, 3D approaches, which have been implemented in 25 studies, are gaining popularity as a result of their capacity to capture volumetric information, which has the potential to result in more precise pCT generation. The 2.5D approach, a compromise between 2D and 3D methodologies, is less common, with only 4 studies using this method. A few studies did not state their approach. The choice of training methodology can significantly impact the performance and accuracy of pCT generation. 2D models may be limited in capturing the full spatial context of MRI data, whereas 3D models, despite being more resource-intensive, can provide a more comprehensive representation of anatomical structures. The 2.5D approach attempts to balance these aspects by considering slices from multiple planes. In synthetic CT generation, the choice between 3D and 2.5D training methodologies impacts the quality and efficiency of the resulting images. 3D methodologies capture full volumetric information, providing a complete anatomical

representation that enhances accuracy in imaging but requires higher computational resources. In contrast, 2.5D methodologies balance spatial detail and computational efficiency by using information from multiple planes, achieving substantial spatial awareness without the intensive demands of full 3D processing. Studies have shown that both approaches are effective: 3D methodologies offer detailed anatomical context, while 2.5D provides a computationally efficient alternative. The dimensionality of training can also influence outcomes, affecting accuracy, reliability, and other primary objectives of synthetic CT generation in radiotherapy planning. The field of pCT generation from MRI is constantly evolving due to variability in training methodologies, necessitating further research and comparative studies for standardized protocols.

Types of Deep Learning Models Used

Authors or researchers implemented various algorithms to produce synthetic CT images from MRI data. The U-Net model was the most frequently used model, as it was utilized in 34 articles. This was primarily due to its capacity to capture fine details and structures through its symmetric encoder-decoder architecture and skipped connections.^{15,27-29,36,38-66} Conditional GANs (cGANs) was also featured in 24 articles, this model conditioned both the generator and discriminator on additional information which enhances control over the generated images.^{32,34,35,42,46,47,55,56,63,67-81} Cycle-GAN, which is recognized for its ability to manage unpaired image-to-image translation by guaranteeing that the translated image can be remapped to the original using cycle consistency loss, was also employed in 19 articles.^{33,40,42,43,57,67,73,74,78, 82-91} 12 articles employed Pix2pix, an additional GAN variant that was developed for paired image-to-image translation tasks.^{40,43,63,71-73,75,77,88,91-93} In 11 articles,

dilated convolutional neural networks (DCNNs) were employed to expand the receptive field without sacrificing resolution, thereby capturing a broader range of contextual information.^{27,45,58,87,94-99} Generative Adversarial Networks (GANs) were utilized in 10 articles, leveraging their two-network system comprising a generator and a discriminator to refine image quality through adversarial training.^{15,30,31,42,89,100-104} Residual network (ResNet) appeared in 10 articles, facilitating the training of very deep networks by addressing the vanishing gradient problem.^{31,39,44,55,57,63,76,86,88,101} Two articles each featured Deep Convolutional Encoder-Decoder Networks (DECNNs) and Fully Convolutional Networks (FCNs). These networks are known for their end-to-end learning capability for pixel-wise classification tasks and are designed to learn efficient representations and reconstruct input data for detailed image reconstruction tasks.^{30,38,104,105} Additionally, a Multichannel Denoising Diffusion Probabilistic Model (MC-DDPM)¹⁰⁶ was utilized in 1 article, and 1 article did not state the specific deep learning model used¹⁰⁷. **Figure 3** illustrates the trends in the number of articles using different deep learning models for synthetic CT (sCT) generation from MRI data over time. The U-Net model shows a significant increase in usage, peaking in recent years, indicating its widespread adoption and effectiveness. The cGAN model also demonstrates a strong presence, with a notable rise in the number of articles over time. Other models like Cycle-GAN, pix2pix, and ResNet have moderate representation, showing steady usage. The minimal usage of less common models, such as MC-DDPM and DECNN, is indicative of their niche applications. The expanding interest and progress in the field of medical imaging are illustrated by this visualization, which emphasizes the changing landscape of deep learning applications.

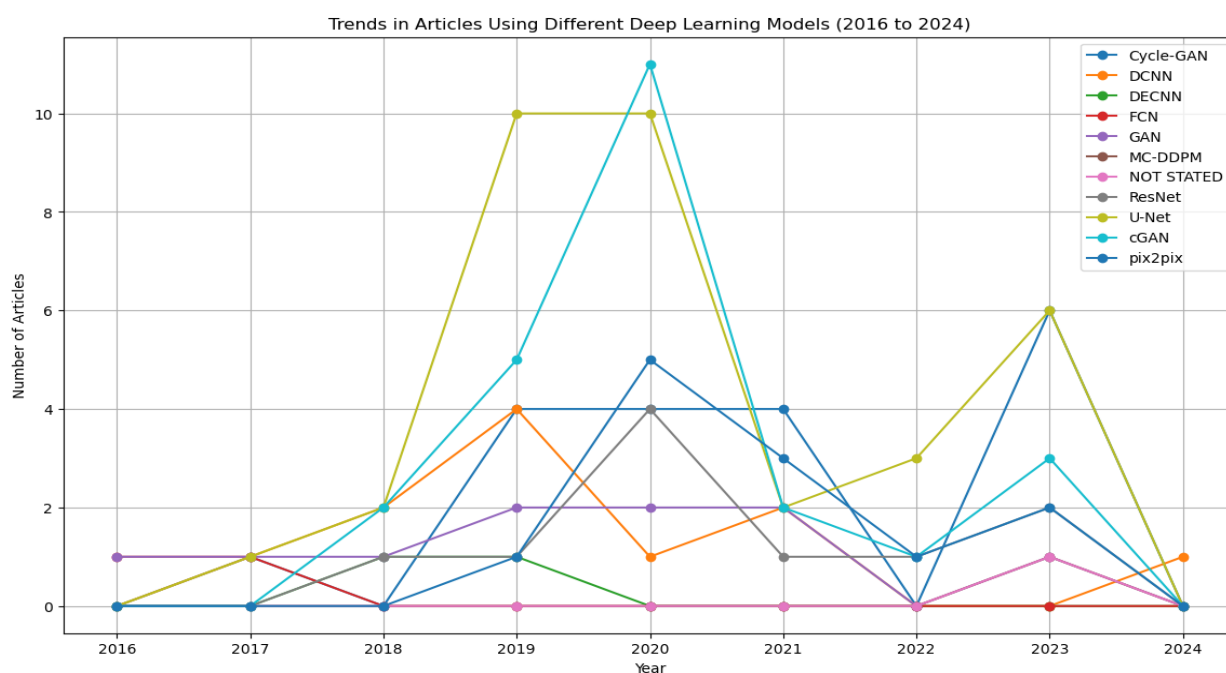


Figure 3. Trends in articles using different deep learning models (2016-2024)

Data preprocessing techniques

MRI sequences used

This section offers a comprehensive description of the diverse MRI sequences that have been implemented in numerous studies to produce synthetic CT (sCT) images from MRI data. Some of the most frequently encountered sequences are T1-weighted, T2-weighted, and FLAIR (Fluid-Attenuated Inversion Recovery). T1-weighted sequences are frequently used (38 articles) due to their high spatial resolution and ability to provide detailed anatomical information, making them ideal for differentiating between different tissue types. T2-weighted sequences were used in 22 articles, T2-weighted sequences are useful for highlighting fluid differences in tissues, providing contrast between normal and pathological areas. Four articles utilized FLAIR sequences, which are particularly advantageous in brain imaging due to their ability to reduce the signal from cerebrospinal fluid, thereby improving the visibility of lesions. Other sequences like post-gadolinium T1-weighted and IDEAL (Iterative Decomposition of Water and Fat with Echo Asymmetry and Least-squares estimation) are used for specific applications, such as contrast-enhanced imaging and fat-water separation. The choice of MRI sequence varies by treatment site, with T1-weighted sequences commonly used for brain imaging and T2-weighted sequences preferred for the pelvis. These diverse MRI sequences enable comprehensive imaging and better synthetic CT (sCT) generation, catering to various clinical needs and improving the accuracy of synthetic CT images.

Preprocessing steps

Preprocessing is a crucial step in generating high-quality synthetic CT (sCT) images by reducing noise, correcting inhomogeneities, normalizing intensities, aligning and registering images, and selecting robust features.¹⁰⁸ In this review, Rigid Registration (RR), used in 47 articles,^{15,27-29,32,33,38,39,43-46,54,56-58,61,63,64,66-68,70,71,75,79,80,83-86,88,91,92,95,96,98,100-102,105-107,109-112} is crucial for aligning MRI and CT images accurately, ensuring that anatomical structures correspond correctly. Deformable Registration (DR), employed in 38 articles,^{15,20,35,36,41,45,47-53,55,58-60,62-64,73,74,76-78,80,81,87,88,90,94,97-99,102,113,114} allows for more flexible alignment, accommodating variations in patient anatomy. Registration techniques are similarly selected according to the treatment site, with rigid registration often applied in brain imaging and deformable registration used for the pelvis to accommodate anatomical variability. Intensity Normalization, used in 24 articles,^{15,20,31,34,43,44,46,56,57,62,64,65,67,70,73,74,79,85,92,93,96,106,109,111} is essential for standardizing image intensity values across different scans, improving consistency and comparability. N4 Bias Field Correction, used in 14 articles,^{34,35,39,54,56,62,63,68,75,78,87,91,99,102} and N3 Bias Field Correction, used in 4 articles,^{27,58,67,105} both correct for intensity inhomogeneities, enhancing image quality. Cropping and Resampling, applied in 13 articles,^{20,48,50,66,84-86,88,95,101,106,107,111} prepare images by adjusting their size and

resolution. Histogram Matching, used in 10 articles, adjusts the intensity distributions of images to match a reference, improving similarity. Geometry Correction, used in 5 articles,^{34,36,84,109,110} corrects for distortions in MRI images. Data Augmentation and Intensity Clipping, each used in 3 articles,^{57,64,114} enhance the training dataset by creating variations and limiting intensity ranges, respectively. Each technique is listed with references to the articles, indicating the diverse preprocessing methods applied in the field of synthetic CT (sCT) generation from MRI using deep learning algorithms.

Loss functions and optimization

Various loss functions are employed to optimize these deep learning models for synthetic CT (sCT) generation. L1 loss, which measures the mean absolute error between predicted and actual values, promotes pixel-wise accuracy and is commonly used in U-Net and DECNN architectures. L2 loss, measuring mean squared error, penalizes larger errors more heavily and is used in models like GANs and DECNN. Adversarial loss, central to GAN-based models, ensures the generated images are indistinguishable from real images by refining the generator's output through adversarial training. Cycle consistency loss, used in Cycle-GAN models, maintains consistency between source and target domains by ensuring that the translated image can be mapped back to the original. Loss functions significantly enhance the accuracy and quality of synthetic CT (sCT) images, making deep learning models highly effective for this application.

Evaluation metrics

Image fidelity metrics

The performance of synthetic CT (sCT) (pCT) generation from MRI using deep learning algorithms is significantly variable across various anatomical regions and studies, as revealed by the systematic review. For brain MRI, Mean Absolute Error (MAE) values range widely, such as 84.8 ± 17.3 ,²⁷ 92.5 ± 13.9 ,³⁰ and 55.7 ± 9.4 ⁶⁷ as shown in **Table 1**, while for head and neck MRI, MAE values include 75 ± 9 ,³⁶ 92.4 ± 13.5 ,⁷³ and 131 ± 24 .⁴⁵ Breast, liver, and abdomen MRI studies report MAE values such as $\text{Body} = 72.9 \pm 18.2$ ⁸⁷ and $\text{Body} = 57 \pm 12$ ⁵³ as presented in **Table 1**. Pelvis MRI studies show MAE values like 42.4 ± 5.1 ¹⁰⁴ and 30.0 ± 4.9 .⁵⁹ Peak Signal-to-Noise Ratio (PSNR) values, which indicate image quality, also vary significantly. For instance, Nie et al. (2017) reported 27.6 ± 1.3 dB, Xiang et al. (2018) 27.3 ± 1.1 dB, and M. C. Florkow et al. (2020) 30.3 ± 1.6 dB.^{30,53,105} In pelvis MRI, Nie et al. (2016) reported a PSNR of 33.4 ± 1.1 dB. Structural Similarity Index (SSIM) values, which measure the structural similarity between synthetic and real CT images, also show considerable variability. Lei et al. (2019) reported a high SSIM of 0.96 ± 0.01 for brain MRI, while Qi et al. (2020) noted $T1 = 0.84 \pm 0.02$ and $T2 = 0.78 \pm 0.03$ for head and neck MRI. Olberg et al. (2019) reported an exceptionally

high SSIM of 0.9995 ± 0.0003 for breast MRI, and Bahrami et al. (2020) reported an SSIM of 0.96 for pelvis MRI.^{46,52,62,67} These metrics underscore the variability in pCT generation accuracy and highlight the need for continuous improvement and validation of deep learning methods in different clinical contexts.

Geometric fidelity results

The data includes Dice Similarity Coefficient (DSC) values from a variety of studies that have utilized deep learning algorithms to generate synthetic CT (sCT) images from MRI images in multiple anatomical regions, such as the brain, head and neck, breast, liver, abdomen, and pelvis. In calculating the Dice Score, structures were delineated either on the CT and MRI or on the CT and sCT, depending on the study. The delineation approach which was either manual, automatic, or segmentational varied across studies. This metric assesses both

the fidelity of the synthetic CT (sCT) in reproducing anatomical structures and the quality of image registration between the modalities. In brain MRI studies, soft tissues and body contours exhibited high DSC values (up to 0.984), while bones and air yielded moderate results. (as low as 0.69) as indicated in **Table 2**.^{32,38,41,67,70,83,94,95,101,109} The DSC for bones and air was moderate to high in head and neck MRI studies, with variability across different sequences.^{36,46,75,98} Breast, liver, and abdomen MRI studies reported DSC values highlighting high accuracy for lungs (up to 0.92) and moderate for bones and gastrointestinal air.^{51,53,57} Pelvis MRI studies demonstrated high DSC values for bones in several studies (up to 0.93), indicating reliable bone segmentation.^{53,58,64,99} The data as shown in **Table 2**, in general, indicates that deep learning algorithms are effective in generating synthetic CT (sCT) images from MRI. However, the results are subject to variation based on the tissue type and anatomical region, indicating that there are areas for further research and improvement.

Table 1. Summary of Image Fidelity Metrics across Various Studies (continued on next page).

ARTICLES	MAE	MSE	PSNR	SSIM	NCC
[27]	84.8 ± 17.3	188.6 ± 33.7	–	–	–
[30]	92.5 ± 13.9	–	27.6 ± 1.3	–	–
[33]	73.7 ± 2.3	–	32.5 ± 0.7	–	–
[109]	Body = 67 ± 11 Soft tissue = 22 ± 3 Bone = 174 ± 29 Air = 159 ± 22	–	–	–	–
[105]	85.4 ± 9.2	–	27.3 ± 1.1	–	–
[34]	129	–	24.2	0.78	–
[28]	All voxels = 81.0 ± 14.6 Soft tissue = 17.6 ± 3.4 Bone = 193.1 ± 38.3 Air = 233.8 ± 28.0	–	–	–	–
[32]	47.2 ± 11.0 60.2 ± 22.0	–	–	–	–
[35]	<i>T1w sequence</i> Body = 120.1 ± 20.4 Soft tissue = 46.3 ± 9.3 Bone = 399.4 ± 51.8 <i>All sequences</i> Body = 108.1 ± 24.0 Soft tissue = 38.9 ± 10.7 Bone = 366.2 ± 62.0	–	–	–	–
[67]	55.7 ± 9.4 57.7 ± 8.4	–	26.6 ± 2.3 27.0 ± 2.8	–	0.96 ± 0.01 0.96 ± 0.01
[94]	75 ± 23	–	–	–	–
[29]	55 ± 10 90 ± 20	–	–	–	–
[68]	54.6 ± 6.8	–	–	–	0.96 ± 0.01
[38]	Body = 54 ± 7 Cerebrospinal fluid = 10 ± 3 Gray matter = 8 ± 2 White matter = 6 ± 2 Air = 53 ± 32	–	–	–	–
[39]	Head = 81 ± 22 Water = 38 ± 11 Bone = 228 ± 63 Air = 274 ± 63 Head = 90 ± 21	–	–	–	–

Table 1. Summary of Image Fidelity Metrics across Various Studies (*continuation; continued on next page*).

ARTICLES	MAE	MSE	PSNR	SSIM	NCC
[92]	Normal soft tissue = 6 ± 3 Air-bone interface = 237 ± 31	–	–	–	–
[69]	47.2 ± 11.0	–	–	–	–
[110]	75.5 ± 7.9	$4.9 \cdot 10^4 \pm 7.7 \cdot 10^3$	25.4 ± 0.6	0.94 ± 0.01	–
	75.5 ± 11.7	$1.1 \cdot 10^4 \pm 3.1 \cdot 10^3$	32.2 ± 1.1	0.94 ± 0.01	–
	74.2 ± 12.8	$1.0 \cdot 10^4 \pm 3.1 \cdot 10^3$	32.4 ± 1.2	0.94 ± 0.01	–
	94.6 ± 17.2	$1.6 \cdot 10^4 \pm 4.9 \cdot 10^3$	30.3 ± 1.2	0.91 ± 0.01	–
	87.7 ± 7.9	$1.4 \cdot 10^4 \pm 2.1 \cdot 10^3$	30.9 ± 0.6	0.92 ± 0.00	–
	98.7 ± 12.7	$1.7 \cdot 10^4 \pm 3.3 \cdot 10^3$	29.9 ± 0.9	0.91 ± 0.01	–
[70]	61.0 ± 14.1	–	26.7 ± 1.9 dB	0.86 ± 0.03	–
[41]	T1w: Brain = 51.2 ± 4.5 Bone = 31.1 ± 7.0	–	T1w: Brain = 43.0 ± 2.0 Bone = 43.2 ± 1.9	T1w: Brain = 0.65 ± 0.05 Bone = 0.87 ± 0.03	–
	T2-FatSat: Brain = 45.7 ± 8.8 Bone = 30.3 ± 7.1		T2-FatSat: Brain = 43.4 ± 1.2 Bone = 43.7 ± 1.2	T2-FatSat: Brain = 0.63 ± 0.03 Bone = 0.86 ± 0.02	
	T1 Post contrast: Brain = 44.6 ± 7.5 Bone = 30.2 ± 6.0		T1 Post contrast: Brain = 43.4 ± 1.2 Bone = 43.7 ± 1.2	T1 Post contrast: Brain = 0.64 ± 0.03 Bone = 0.86 ± 0.03	
	T2 CUBE FLAIR: Brain = 51.2 ± 4.5 Bone = 36.1 ± 3.3		T2 CUBE FLAIR: Brain = 44.9 ± 1.2 Bone = 45.2 ± 1.1	T2 CUBE FLAIR: Brain = 0.61 ± 0.04 Bone = 0.84 ± 0.02	
[71]	60.5 ± 13.3	–	49.2 ± 1.9	–	–
[20]	Oral cavity: 15.7 ± 4.1 HU Parotid: 23.2 ± 7.1 HU Spinal cord: 11.5 ± 4.1 HU Cavity: 212.8 ± 104.6 HU Mandible: 219.4 ± 35.1 HU Teeth: 268.8 ± 88.8 HU	–	–	0.906 ± 0.019	–
[31]	89.30 ± 10.25	–	26.64 ± 1.17	0.83 ± 0.03	–
[83]	131 ± 14.3 HU (body contour) and 246.3 ± 62.9 HU (bone contour)	–	–	–	–
[95]	62.2 ± 4.1 HU	–	–	–	–
[84]	Adapted model on T1- FLAIR: 74.56 ± 8.61	–	Adapted model on T1- FLAIR: 28.30 ± 0.83	Adapted model on T1- FLAIR: 0.84 ± 0.01	–
	Adapted model on T1- POST: 74.89 ± 15.64		Adapted model on T1- POST: 27.72 ± 1.43	Adapted model on T1- POST: 0.83 ± 0.04	
[101]	75 ± 23 HU	–	–	–	–
[106]	Brain dataset MAE: 43.317 ± 4.104 HU	–	Brain dataset PSNR: 27.046 ± 0.817 dB	Brain dataset SSIM: 0.965 ± 0.005	Brain dataset NCC: 0.983 ± 0.004
	Prostate dataset: MAE: 59.953 ± 12.462 HU	–	Prostate dataset: PSNR: 26.920 ± 2.429 dB	Prostate dataset: SSIM: 0.849 ± 0.041	Prostate dataset: NCC: 0.948 ± 0.018
[107]	–	–	–	–	–
[85]	65.3 ± 13.9 HU	–	28.5 ± 2.2 dB	0.90 ± 0.03	–
[111]	–	0.00118 ± 0.00037	29.47 ± 1.35 dB	0.84 ± 0.036	–
[43]	U-Net: 26.8 ± 10.0 HU Pix2pix: 24.3 ± 8.0 HU CycleGAN: 23.8 ± 8.6 HU (compared to 42.2 ± 17.4 HU for original CBCT)	–	U-Net: 29.1 ± 1.7 dB Pix2pix: 31.3 ± 1.9 dB CycleGAN: 37.8 ± 2.1 dB (compared to 27.2 ± 1.9 dB for original CBCT)	U-Net: 0.94 ± 0.01 Pix2pix: 0.95 ± 0.01 CycleGAN: 0.96 ± 0.01 (compared to 0.91 ± 0.03 for original CBCT)	–
[44]	Body T1: 79.8 ± 5.9 HU T2: 71.1 ± 3.1 HU T1CM: 82.9 ± 6.1 HU	–	–	T1: 0.97 ± 0.01 T2: 0.98 ± 0.00 T1CM: 0.96 ± 0.01	–
	Bone: T1: 216.3 ± 29.6 HU T2: 186.1 ± 25.7 HU T1CM: 236.4 ± 41.4 HU				
[36]	Body = 75 ± 9	–	–	–	–

Table 1. Summary of Image Fidelity Metrics across Various Studies (*continuation*).

ARTICLES	MAE	MSE	PSNR	SSIM	NCC
[73]	Body = 92.4 ± 13.5 Body = 100.7 ± 14.6	–	–	–	–
[45]	131 ± 24	–	–	–	–
[102]	82.8 ± 48.6	–	–	–	–
[74]	Body = 69.7 ± 9.3 Body = 100.6 ± 7.7	–	–	–	–
[46]	Body: T1 = 75.2 ± 11.5 T2 = 87.0 ± 10.8 T1C = 80.0 ± 10.9 T1DixonC = 86.3 ± 10.8 Multiseq. = 70.0 ± 12.0 71.3 ± 12.4	–	T1 = 28.9 ± 1.1 T2 = 27.5 ± 0.9 T1C = 28.4 ± 1.1 T1DixonC = 27.7 ± 0.9 Multiseq. = 29.4 ± 1.3 29.2 ± 1.3	T1 = 0.84 ± 0.02 T2 = 0.78 ± 0.03 T1C = 0.82 ± 0.03 T1DixonC = 0.79 ± 0.03 Multiseq. = 0.85 ± 0.03 –	–
[97]	Body = 65.4 ± 3.6	–	–	–	–
[75]	Head = 75.7 ± 14.6	–	29.1 ± 1.6	0.92 ± 0.02	–
[98]	Body = 67 ± 14	–	–	–	–
[47]	MAE (Mean Absolute Error) between pCT and CBCT: 180.82 ± 27.37 HU MAE for DRUnet: 67.88 ± 8.39 HU MAE for cGAN: 72.52 ± 8.43 HU MAE for MPBM: 90.69 ± 14.3 HU	–	–	CBCT: 0.88 ± 0.03 DRUnet: 0.95 ± 0.01 cGAN: 0.94 ± 0.01 MPBM: 0.93 ± 0.02	–
[66]	Light_U-Net: 95.6 HU	Light_U-Net: 54.3×10^3 HU	–	–	–
[48]	CBCT: 44.38 HU sCTU-net (best configuration with pCT): 18.98 HU	–	CBCT: 27.35 dB sCTU-net (best configuration with pCT): 33.26 dB	CBCT: 0.7109 sCTU-net (best configuration with pCT): 0.8911	–
[49]	CBCT: 49.90 HU USsCT: 28.52 HU	–	CBCT: 26.74 dB USsCT: 30.75 dB	CBCT: 0.9079 USsCT: 0.9642	–
[103]	32.70 HU	–	34.12 dB	0.86	–
[87]	Body = 72.9 ± 18.2	–	22.4 ± 3.6	–	–
[52]	–	–	71.7 ± 2.3	0.9995 ± 0.0003	–
[76]	60.4	–	–	–	–
[77]	Body = 78.7 ± 18.5	–	–	–	–
[114]	Body = 57 ± 12 Lungs = 105 ± 34	–	Body = 30.3 ± 1.6	–	–
[78]	Body = 89.8 ± 18.7 Body = 94.1 ± 30	–	27.4 ± 1.6 27.2 ± 2.2	–	–
[54]	Liver = 24.1 ± 11.5 Kidneys = 47.1 ± 15 Spinal Cord = 29.8 ± 9.4 Lungs = 105.7 ± 35.0	–	–	–	–
[55]	55.6 ± 5.6	–	–	–	–
[56]	body: 57.08 ± 2.79 HU Air (54.42 ± 11.48 HU), Bone (86.03 ± 10.76 HU), Soft tissues (55.39 ± 3.41 HU)	–	27.64 ± 0.68 dB	–	0.92 ± 0.02
[57]	Mean Absolute Error (MAE): 55.0 ± 16.6 HU (proposed) vs. 58.9 ± 16.8 HU (baseline)	–	–	–	Normalized Cross- Correlation (NCC): 0.97 ± 0.02 (proposed) vs. 0.96 ± 0.02 (baseline)

Table 2. Summary of Dice Similarity Coefficient (DSC) Values for Different Anatomical Structures in Various Studies .

ARTICLES	TREATMENT SITE	DICE SIMILARITY COEFFICIENT
[109]	Head and Neck	Body contour = 0.98 ± 0.01 Bone = 0.85 ± 0.04 Air = 0.71 ± 0.07
[32]	Brain	Head = 0.96 ± 0.02 Bone = 0.80 ± 0.06 Air = 0.70 ± 0.07
[67]	Brain, Pelvis	Air = 0.90 ± 0.12 Bone = 0.83 ± 0.06
[93]	Prostate	Soft tissue = 0.94 ± 0.02 Bone = 0.85 ± 0.02 Air = 0.95 ± 0.01
[38]	Brain	Bone = 0.93 ± 0.02 Air = 0.92 ± 0.03
[70]	Brain	Body = 0.984 ± 0.004 Bone = 0.92 ± 0.05
[41]	Brain	T1w: Soft tissue = 0.91 ± 0.03 Bone = 0.76 ± 0.12
		T2-FatSat: Soft tissue = 0.91 ± 0.02 Bone = 0.77 ± 0.07
		T1 Post contrast: Soft tissue = 0.90 ± 0.02 Bone = 0.75 ± 0.07
		T2 CUBE FLAIR: Soft tissue = 0.88 ± 0.03 Bone = 0.69 ± 0.07
		$77.2 \pm 3.3\%$ for bone contour Sensitivity: $90.5 \pm 1.2\%$ Specificity: $82.2 \pm 5.6\%$
[83]	Brain	
[95]	Brain	bone overlap between sCT and CT images: 0.92 ± 0.01
[101]	Head and Neck	Dice coefficients (Air: 0.95, Soft tissue: 0.94, Bone: 0.85),
[36]		Bone = 0.70 ± 0.07 Air = 0.79 ± 0.08
[46]	Head and Neck	Bone: T1 = 0.74 ± 0.05 T2 = 0.68 ± 0.05 T1C = 0.72 ± 0.05 T1DixonC = 0.69 ± 0.06 Multiseq. = 0.77 ± 0.05 0.76 ± 0.05
[75]	Head and Neck	Bone = 0.86 ± 0.03
[98]	Head and Neck	Bones = 0.80 ± 0.07 Air = 0.81 ± 0.1
[51]	Breast	Patient 1 = 76.4 Patient 2 = 73.5
[114]	Abdomen	Bones = 76 ± 8 Lungs = 92 ± 9
[57]	Abdomen	Dice Similarity Coefficient (DSC) for GI air: 0.872 ± 0.053 (proposed) vs. 0.846 ± 0.052 (baseline) DSC for skeleton: 0.862 ± 0.020 (proposed) vs. 0.849 ± 0.024 (baseline) DSC for soft tissues: 0.974 ± 0.007 (proposed) vs. 0.971 ± 0.007 (baseline)
[58]	Pelvis	Bone = 0.93 ± 0.02
[60]	Pelvis	Bone = 0.89 ± 0.02
[99]	Pelvis	Bone = 0.81 ± 0.04
		Bone = 0.82 ± 0.04
		Bone = 0.80 ± 0.05
[67]	Brain, Pelvis	Air = 0.75 ± 0.06 Bone = 0.81 ± 0.05
[112]	Abdomen	Bone = 0.81 ± 0.06
		Bone = 0.81 ± 0.06
		Bone = 0.85 ± 0.05
[62]	Pelvis	Bone = 0.77 ± 0.03 Bone = 0.7 ± 0.09
[64]	Pelvis	0.85 ± 0.05

Dosimetric evaluation

The use of deep learning algorithms to generate synthetic CT (sCT) from MRI data has been shown to be highly accurate in a variety of body regions, with dose differences typically below 1 percent and high gamma pass rates, as confirmed by the analysis of the data from various studies in **Table 3**. In brain MRI studies, dose differences were generally under 1%, with examples such as Dinkla et al. (2018) and Kazemifar et al. (2019) reporting <1% and <2.4%, respectively. High gamma pass rates were observed, such as 3%/3mm = 99.3% in Dinkla et al. (2018) and 2%/2mm = 99.2% in Kazemifar et al. (2019). Head and neck MRI studies also showed minimal dose differences, with Dinkla et al. (2019) and Klages et al. (2019) reporting <1% and

PTV70Gy < 0.8%, respectively, and high gamma pass rates such as 2%/2mm = 98.7% in Peng et al. (2020). For breast, liver, and abdomen MRI, studies like Y. Liu et al. (2019) and Olberg et al. (2019) reported dose differences of <1 Gy and <0.9%, respectively, with gamma pass rates like 2%/2mm = 97.0% in Y. Liu et al. (2019). Pelvis MRI studies, including those by Arabi et al. (2018) and Chen et al. (2018), reported dose differences of <0.5% and 0.2%, with high gamma pass rates such as 2%/2mm = 98.5% in Arabi et al. (2018) and 2%/2mm = 99.4% in Chen et al. (2018). Overall, the data presented in **Table 3** consistently support the reliability and clinical potential of synthetic CT (sCT) generation from MRI in radiotherapy planning.

Table 3. Comparative Analysis of Dose Differences and Gamma Pass Rates in Various Studies (continued on next page).

ARTICLE	DOSE DIFFERENCES (Mean Error, ME)	GAMMA PASS RATES
[109]	< 1%	The gamma pass-rates were 91.1% for 1%/1 mm, 95.8% for 2%/2 mm, and 99.3% for 3%/3 mm.
[32]	< 2.4% for PTV and OARs	The gamma pass-rates were 94.6% for 1%/1 mm and 99.2% for 3%/3 mm.
[35]	< 1%	For the T1-weighted (T1w) sequence, the gamma pass-rates are 94.2% for 1%/1 mm, 98.9% for 2%/2 mm, and 99.7% for 3%/3 mm. For all sequences, the gamma pass-rates are 95.3% for 1%/1 mm, 99.2% for 2%/2 mm, and 99.8% for 3%/3 mm.
[94]	< 1.5%	The gamma pass-rates is 99.2% for 3%/3 mm.
[29]	–	The gamma pass-rates are 95% for 1%/1 mm, 98% for 2%/2 mm, and 91.7% for 3%/3 mm. The gamma pass-rates are 95% for 1%/1 mm, 98% for 2%/2 mm, and 89.3% for 3%/3 mm.
[68]	< 0.5%	The gamma pass-rates are 89.2% for 1%/1 mm, 98.1% for 2%/2 mm, and 99.9% for 3%/3 mm.
[38]	< 0.1 Gy	–
[36]	< 1%	The 2%/2 mm gamma pass-rate is 95.6%, and the 3%/3 mm gamma pass-rate is 98.7%.
[39]	< 0.3%	The gamma pass-rates are 97.9% for 1%/1 mm, 99.6% for 2%/2 mm, and 99.8% for 3%/3 mm. The gamma pass-rates is 99.7% for 3%/3 mm.
[69]	< 1.8%	–
[110]	–	–
[100]	< 0.13 Gy	The gamma pass-rates are 99.0% for 1%/1 mm and 99.9% for 2%/2 mm.
[70]	< 0.9%	The gamma pass-rate is greater than 93.9% for 2%/2 mm and greater than 98.4% for 3%/3 mm.
	< 1%	The gamma pass-rate is greater than 92.6% for 2%/2 mm and greater than 97.2% for 3%/3 mm.
[41]	–	–
[71]	PTV < 0.13% Brainstem, optic chiasma, and optic nerve < 0.77%	The gamma pass-rates are 97.3% for 2%/2 mm and 99.8% for 3%/3 mm.
[93]	< 0.4 Gy	The 2%/2 mm gamma pass-rate is 99.1%
[95]	below 0.2%	>99.1%
[44]	The dose difference between original proton treatment plans and recalculated plans on sCTs was less than 1% for all input sequences. The dose differences for brainstem and optical nerves varied by $\pm 3.5\%$ across all sequences.	–
[73]	PTV _{70Gy} < 0.8% OARs < 2% PTV _{70Gy} < 1.6% OARs < 1.7%	–
[45]	< 1%	–

Table 3. Comparative Analysis of Dose Differences and Gamma Pass Rates in Various Studies (continuation).

ARTICLE	DOSE DIFFERENCES (Mean Error, ME)	GAMMA PASS RATES
[74]	PTV ($D_{95\%}$) < 0.7% Parotids (D_{mean}) < 0.9% Mandible (D_{max}) < 1.5% PTV ($D_{95\%}$) < 0.9% OARs (D_{mean}) < 1.2% Mandible (D_{max}) < 1.6%	The 2%/2 mm gamma pass-rate is 98.7%, and the 3%/3 mm gamma pass-rate is 99.6%. The 2%/2 mm gamma pass-rate is 98.5%, and the 3%/3 mm gamma pass-rate is 99.6%.
[46]	< 1% < 0.9%	For the T1 sequence, the 2%/2 mm gamma pass-rate is 97.4%; for the T2 sequence, it is 95.8%; for the T1C sequence, it is 96.7%; for the T1DixonC sequence, it is 96.2%; and for the multisequence analysis, it is 97.8%. The gamma pass-rate for 2%/2 mm is 97.6%.
[97]	–	The gamma pass-rate for 2%/2 mm is 97.6%.
[75]	–	–
[98]	PTV and OARs < 0.3%	The gamma pass-rate for 2%/1 mm is 99.4%.
[50]	The study found minor dose differences for target volumes, mostly within $\pm 1.5\%$, with larger dose differences observed in OARs, especially in the left parotid gland.	–
[87]	The difference in PTV _{45Gy} and bowel dose is less than 1 Gy, and the PTV $D_{95\%}$ is less than 1.1%.	The gamma pass-rates are 90.8% for 1%/1 mm, 97.0% for 2%/2 mm, and 99.4% for 3%/3 mm.
[52]	< 0.9%	The gamma pass-rate for 2%/2 mm is $\geq 98.0\%$.
[77]	–	The gamma pass-rate is 90.8% for 1%/1 mm, 98.7% for 2%/2 mm, and 99.8% for 3%/3 mm.
[114]	<0.5% –	For photon therapy, the 2%/2 mm gamma pass-rate is greater than 99%. For photon therapy, the 2%/2 mm gamma pass-rate is greater than 96%.
[78]	< 0.6% < 1%	The 2%/2 mm gamma pass-rate is 97.4%, and the 3%/3 mm gamma pass-rate is 99.5%. The 2%/2 mm gamma pass-rate is 95.6%, and the 3%/3 mm gamma pass-rate is 99.3%.
[54]	< 0.15 Gy	–
[88]	$\pm 2\%$	–
[56]	–	The global gamma pass rate reached 94.88%, meeting the 3%/3 mm criteria.
[58]	Mean dose difference < 0.5% Dose difference: CTV (D_{max}) = 2.9% OARs (D_{max}) = 0.5%	The 1%/1 mm gamma pass-rate is 94.6%, the 2%/2 mm gamma pass-rate is 98.5%, and the 3%/3 mm gamma pass-rate is 99.2%.
[59]	0.2% (PTV/OARs)	The 1%/1 mm gamma pass-rate is 98.0%, the 2%/2 mm gamma pass-rate is 99.4%, and the 3%/3 mm gamma pass-rate is 99.8%.
[79]	For 10 patients /27 Dose difference: PTV ($D_{98\%}$) < 1% For 10 patients /18 Dose difference: PTV ($D_{98\%}$) < 1.2% For 10 patients /14 Dose difference: PTV ($D_{98\%}$) < 1%	For 10 patients out of 27, the gamma pass-rate for 2%/2 mm is 95.0%, and for 3%/3 mm is 98.1% For 10 patients out of 18, the gamma pass-rate for 2%/2 mm is 91.6%, and for 3%/3 mm, it is 97.1%. For 10 patients out of 14, the gamma pass-rate for 2%/2 mm is 90.6%, and for 3%/3 mm, it is 97.1%.
[15]	< 0.6% (PTV and OARs)	The gamma pass-rate for 1%/1 mm is 99.2%.
[112]	0.23%	The gamma pass-rate is 92.4% for 1%/1 mm, 98.0% for 2%/2 mm, and 99.0% for 3%/3 mm.
[80]	PTV Rectum ($D_{95\%}$, $D_{50\%}$ and $D_{2\%}$) < 0.7% PTV Anus ($D_{95\%}$, $D_{50\%}$ and $D_{2\%}$) < 0.5%	The gamma pass-rate is 99.5% for 1%/1 mm, 99.8% for 2%/2 mm, and 100% for 3%/3 mm.
[77]	PTV ($D_{98\%}$, $D_{50\%}$, $D_{2\%}$) < 0.07 Gy Rectum ($D_{98\%}$, $D_{50\%}$, $D_{2\%}$) < 0.05 Gy	The gamma pass-rate is 89.3% for 1%/1 mm, 99.0% for 2%/2 mm, and 99.9% for 3%/3 mm.
[65]	The mean dose differences to the PTV are less than 0.5% for all dummy PTVs and clinical plans, as per rigid registration.	The gamma pass rates were recorded at $96.5 \pm 0.8\%$ in the rectum and $95.4 \pm 0.6\%$ in the anus.

Challenges and future directions

Generating synthetic CT (sCT)s faces challenges like data acquisition complexity, privacy concerns, and diverse datasets for model generalizability.¹⁰ Clinical integration faces challenges due to high computational demands, resource requirements, overfitting, and differences in imaging protocols

across medical centers, hindering generalizability. Future improvements could focus on advanced data augmentation and federated learning to enhance data availability, efficient algorithms and hardware acceleration to reduce computational demands, and advanced transfer learning and domain adaptation techniques to improve generalizability. Robust validation

frameworks and extensive clinical trials are necessary to ensure the reliability and safety of synthetic CT (sCT) images. The successful implementation of synthetic CT (sCT) technologies in healthcare requires interdisciplinary collaboration, integration with other imaging modalities, close clinicians, and ethical and regulatory considerations.

Conclusion

This review underscores the variability in evaluation metrics used across studies, which reflects the challenges in standardizing synthetic CT generation for radiotherapy applications. Despite these challenges, synthetic CT provides significant benefits in radiotherapy planning, offering improved image accuracy and non-invasive mapping capabilities. Further research should prioritize consistency in metrics and continued validation across diverse clinical settings to fully harness the potential of synthetic CT in enhancing radiotherapy outcomes.

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